



StarGANv2 vs Ours

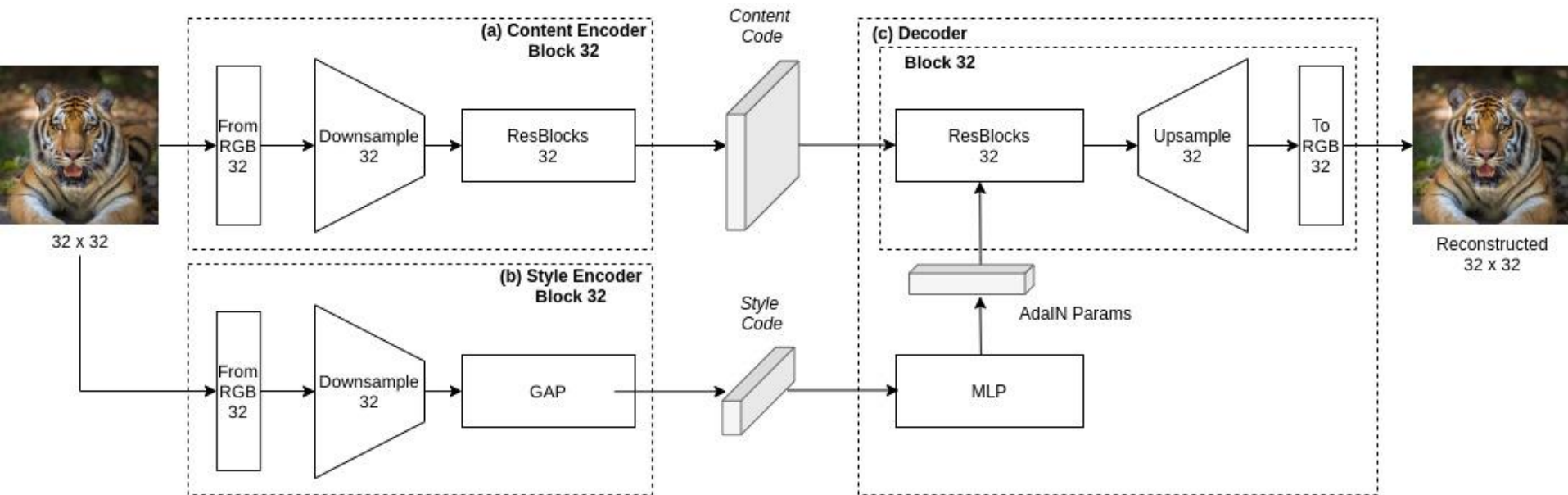
Lab Team



Architecture

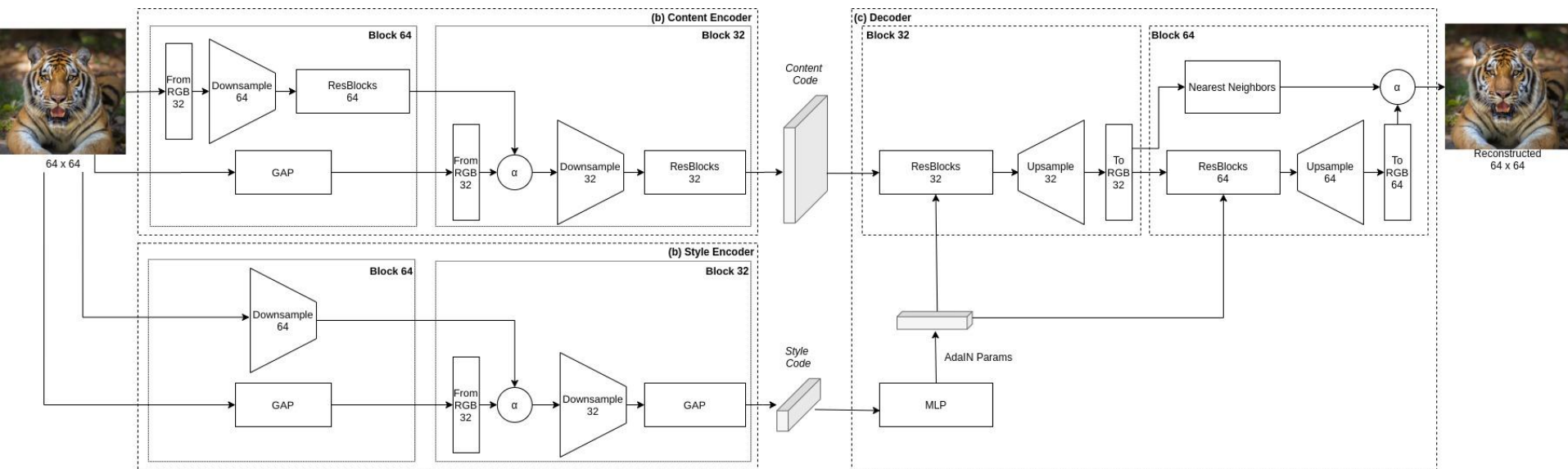
Our model

Encoder - Decoder: Phase 1



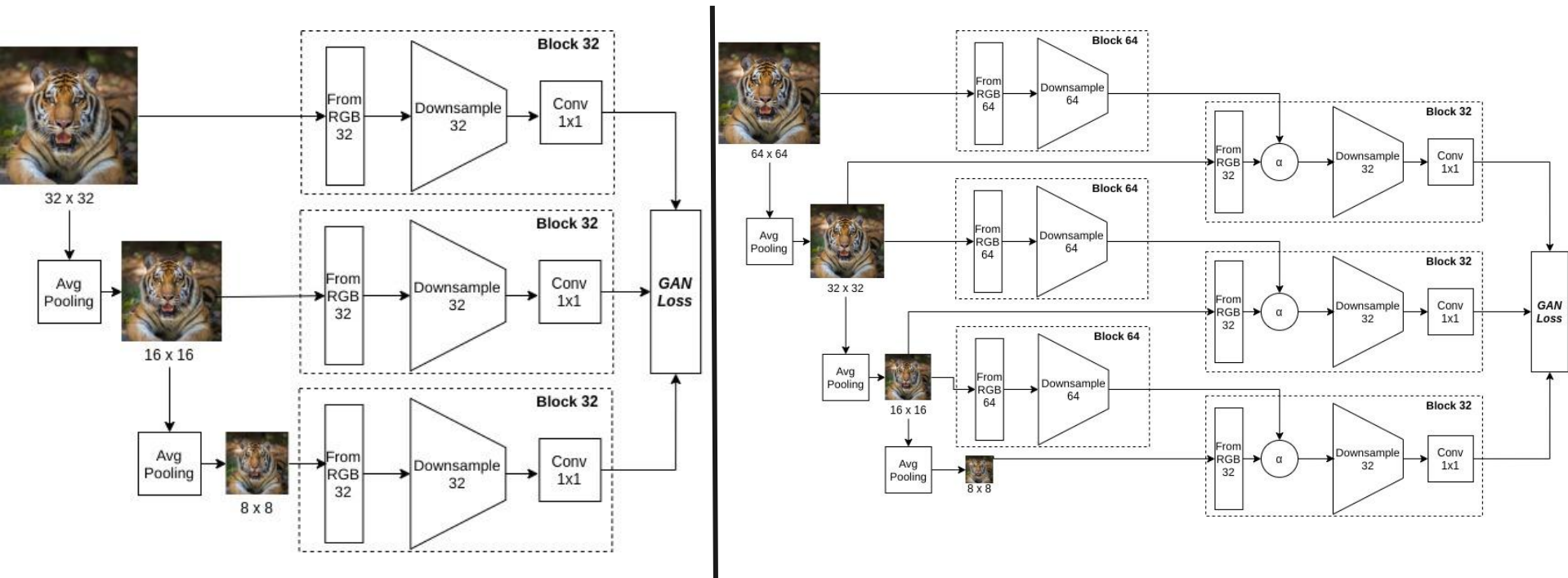
Our model

Encoder - Decoder: Phase 2

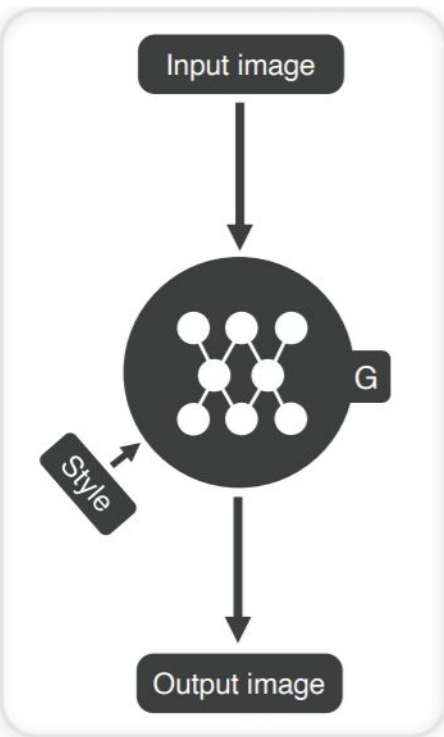


Our model

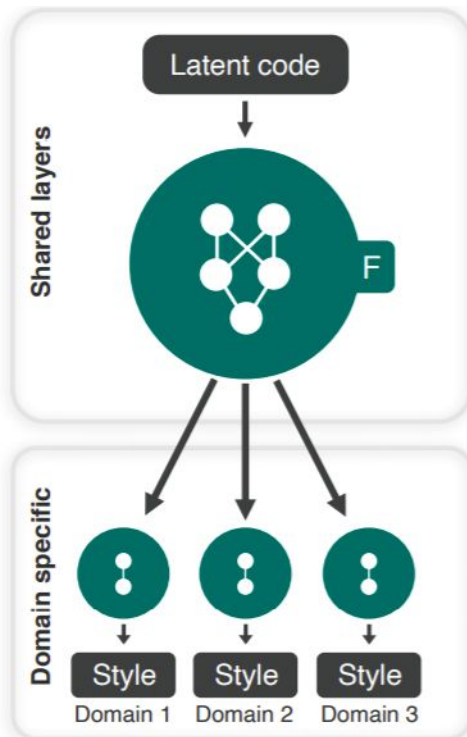
Discriminator: Phase 1 (Left), Phase 2 (Right)



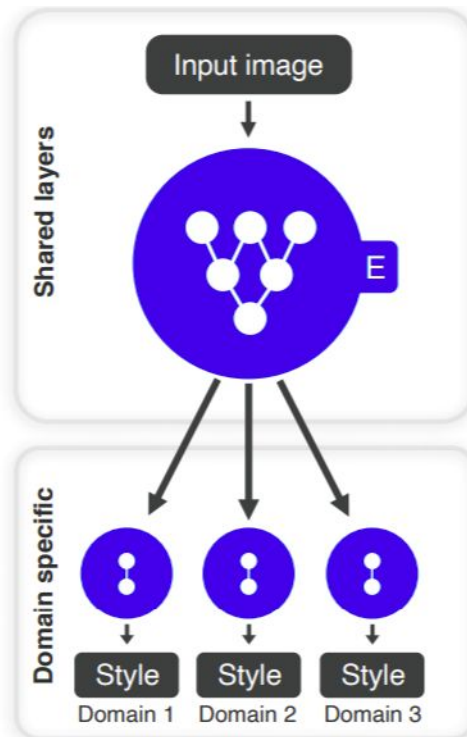
StarGANv2



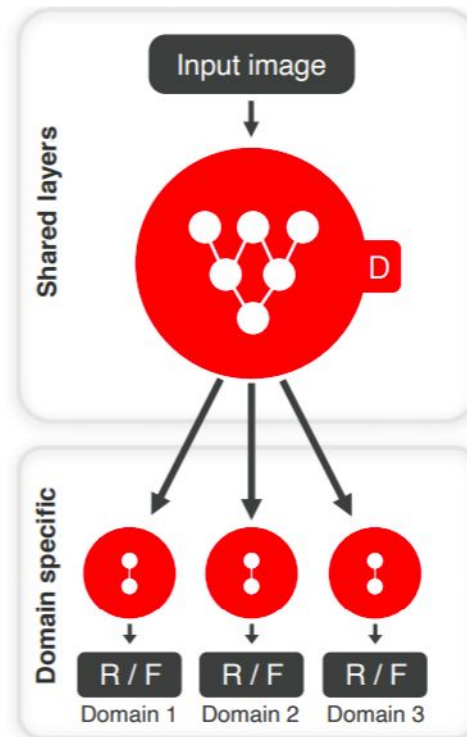
(a) Generator



(b) Mapping network

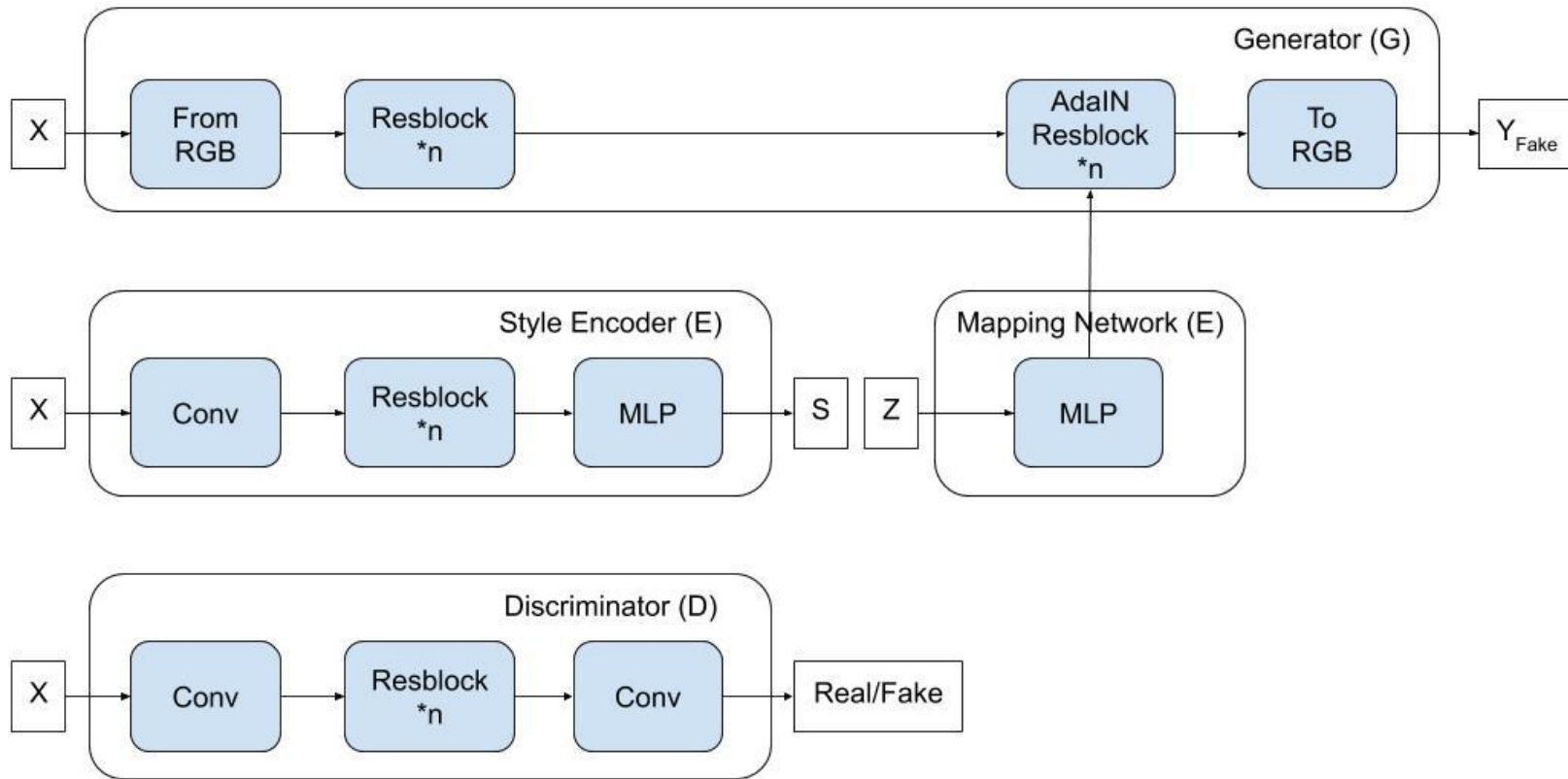


(c) Style encoder



(d) Discriminator

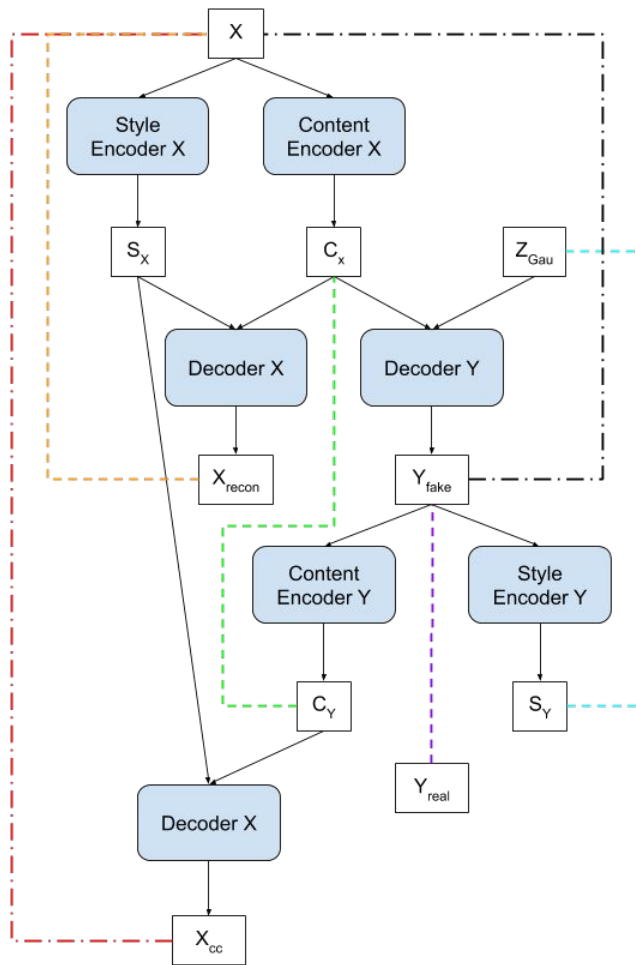
StarGANv2





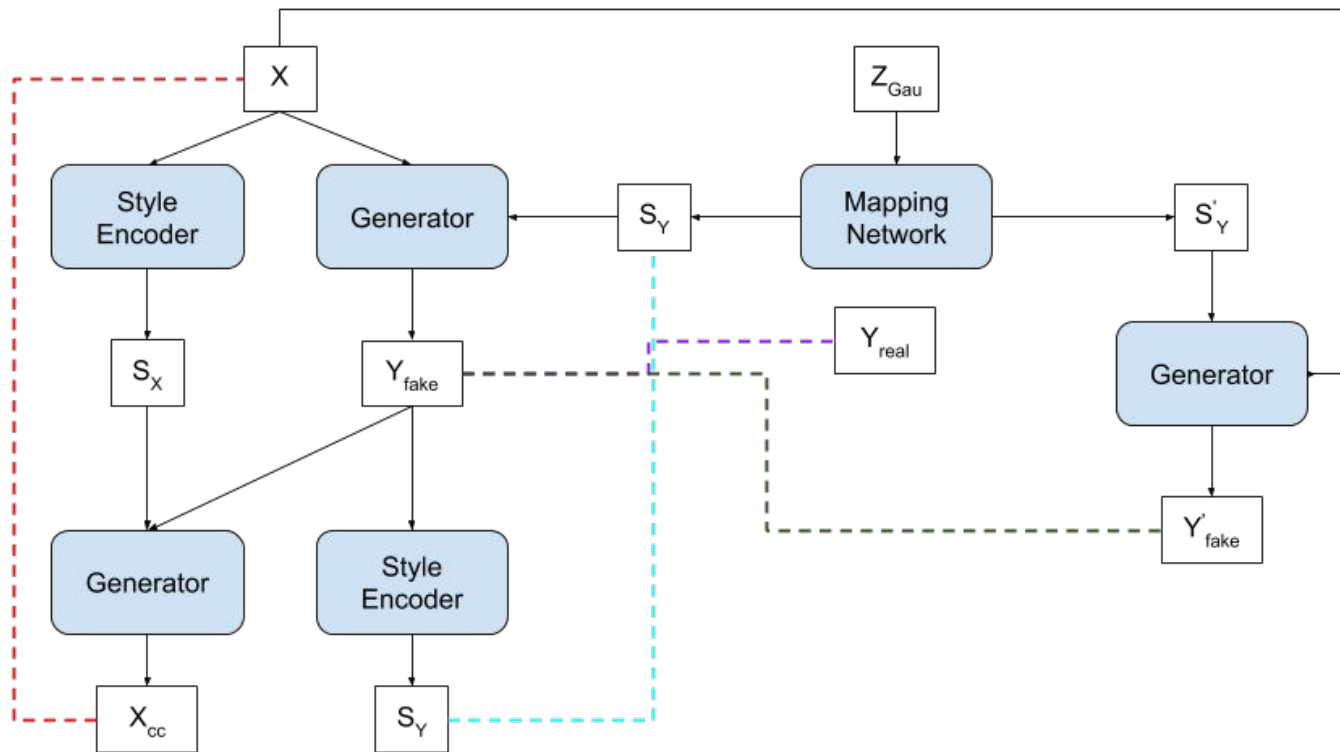
Loss functions

Our model



Cycle consistency loss
Image reconstruction loss
Content reconstruction loss
VGG loss (optional)
Style reconstruction loss
GAN loss

StarGANv2



Cycle consistency loss

Style reconstruction loss

GAN loss

Style diversification



Summary

StarGANv2 vs Ours



StarGANv2	Ours
<i>Shared weights between domains</i>	<i>Each domain has its own weights</i>
<i>Start training from highest resolution</i>	<i>Train progressively</i>
<i>Doesn't use Image reconstruction loss and Content reconstruction loss</i>	<i>Use both Image reconstruction loss ($w=10$) and Content reconstruction loss ($w=1$)</i>
Use R1 regularization (like StyleGAN)	Use LSGAN
Use additional face landmark (with CelebA)	Doesn't use face landmark